

Disentangling Scale from Targeted Pretraining Depth: Testing the Less-Is-More Hypothesis for Bengali Mathematical Reasoning

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Abstract—Bengali is spoken by more than 230 million people and remains thinly represented in language model pretraining. Two recent efforts respond to this by pretraining smaller models heavily on Bengali, on the implicit premise that targeted language depth can stand in for parameter scale. We test that premise directly. Seven backbones from 0.6B to 4B parameters are compared on held-out Bengali mathematical reasoning: four with deep Bengali pretraining (TituLLM and TigerLLM) and three with only incidental Bengali coverage (Qwen3). Each is then fine-tuned with LoRA on 873 curated Bengali reasoning traces, selected in the LIMO difficulty window and decontaminated against three public Bengali mathematics benchmarks, and re-evaluated on the identical items so that every comparison is paired. Scale predicts performance and pretraining depth does not. Accuracy rises from 3.6% to 16.8% to 44.8% across the Qwen3 ladder, while the Bengali-specialised models stay between 0.4% and 4.4% across a threefold parameter range. A 1.7B model with no Bengali continued pretraining outcores a 3.2B model that has it, at every difficulty tier. Minimal-data elicitation moves no model significantly, and a control experiment rules out difficulty miscalibration as the explanation: training on easy, well-matched problems is significantly worse than training on hard ones. Fine-tuning did reshape output format, length and stopping behaviour without improving correctness, which we read as a negative-case instance of the Superficial Alignment Hypothesis. We also quantify a Bengali tokenizer tax of roughly $3.1\times$ and show it is separable from reasoning ability: the two most token-efficient models here are the two weakest at mathematics.

Index Terms—Bengali, low-resource languages, mathematical reasoning, large language models, model scale, continued pretraining, parameter-efficient fine-tuning, tokenizer efficiency

I. INTRODUCTION

Large language models can now solve competition-level mathematics, but nearly all of that progress has happened in English. Bengali has more than 230 million speakers and is still treated as low-resource. It is thinly represented in pretraining corpora, and reasoning-specific Bengali data barely exists. Whether a model can actually reason *in* Bengali, rather than translate its way through a problem, is still an open question.

Two recent efforts approach this gap from opposite directions. TigerLLM (Raihan & Zampieri, 2025) continually

pretrained a 1B-parameter model on roughly 10M tokens of Grade 6–12 Bengali textbook content, plus instruction data distilled from frontier models. It reportedly outperformed much larger generalist systems, which the authors read as evidence that targeted, high-quality pretraining can substitute for scale. TituLLM (Nahin et al., 2025) took the opposite route, pretraining on roughly 37B tokens of general-domain Bengali with an extended tokenizer, and found gains that were capability-specific rather than uniform. Read side by side the two results look contradictory. They are not really a test of the same question, though, because in both cases model size and pretraining design move together. Neither study isolates which factor is doing the work.

That is the first gap this paper addresses. A second follows from it. If reasoning ability can be elicited rather than taught, how much data does that take? The less-is-more literature argues the answer is surprisingly little. LIMA (Zhou et al., 2023) proposed the Superficial Alignment Hypothesis, which holds that a model’s knowledge is acquired almost entirely during pretraining and that alignment mainly teaches it which format to answer in. It demonstrated competitive instruction-following from 1,000 curated examples. LIMO (Ye et al., 2025) extended the claim to mathematical reasoning, and s1 (Muennighoff et al., 2025) reproduced it with 1,000 traces plus inference-time budget forcing. Both LIMO and s1, however, demonstrate their result at 32B parameters and in English. Whether minimal-data elicitation survives the move to a genuinely low-resource language, and to the model sizes most researchers can actually train, has not been tested.

We test both questions with a single controlled experiment. Seven backbones spanning 0.6B to 4B parameters are fine-tuned with LoRA on an identical set of curated Bengali mathematical reasoning traces, then evaluated on a held-out Bengali mathematics benchmark before and after training. The backbones are chosen so that pretraining depth and parameter scale vary separately. TigerLLM-1B and TituLLM-1B match in size but differ by roughly $3,700\times$ in targeted Bengali pretraining. TituLLM-1B and TituLLM-3B share a pretraining recipe and differ in size. The Qwen3 family (0.6B, 1.7B, 4B)

provides a scale ladder with essentially no Bengali-specific pretraining at all.

Our contributions are:

- **A controlled separation of scale from targeted pre-training depth in Bengali.** Within a single model family holding pretraining constant, accuracy rises from 3.6% at 0.6B to 16.8% at 1.7B to 44.8% at 4B, and every adjacent step on that ladder is significant (Fisher exact, $p < 10^{-5}$). Targeted Bengali pretraining produces no comparable gradient: four Bengali-specialised models spanning 1B to 3.2B parameters all sit between 0.4% and 4.4%.
- **No comparison in this study favours the model with more Bengali pretraining.** TigerLLM-1B and TituLLM-1B are matched in size and differ by roughly $3,700\times$ in targeted Bengali pretraining, and the model with less of it scores higher, 4.4% against 2.0%. The sharpest case is TituLLM-3B-Instruct against Qwen3-1.7B: roughly twice the parameters plus a Bengali continued-pretraining corpus, against neither, and it loses 1.2% to 16.8% ($p < 10^{-4}$). This contradicts the substitution claim in (Raihan & Zampieri, 2025), tested on a Bengali benchmark using that work’s own released model.
- **A scale boundary on minimal-data elicitation.** LIMO-style fine-tuning produced no significant accuracy change on any of seven backbones. That includes one model whose 44.8% baseline shows LIMO’s stated precondition of latent capability is clearly satisfied.
- **Evidence that elicitation changes form without changing capability.** No backbone retained even half the items it had answered correctly before tuning, and Qwen3-4B abandoned its own answer format on 222 of 250 items while its accuracy moved by 13. Mean Bengali character adherence fell in six of seven models, by up to 17 points, and the per-item decline is significant in each of those six (exact sign test, $p < 10^{-7}$).
- **A direct test of the difficulty-calibration explanation.** Retraining the same backbone on difficulty-matched easy data performed significantly worse than the hard-data adapter, 12.4% against 18.0% (exact McNemar, $p = 0.039$), which rules out the most obvious alternative account of the null result.
- **A measurement of the Bengali tokenizer tax** and what it means for which models can be trained on consumer hardware.

A. Research Questions

RQ1: When a model is given a small set of Bengali mathematical examples intended to elicit reasoning, what primarily determines its performance: parameter scale, or the amount of Bengali seen during pretraining? We answer this by comparing models that differ in only one of these factors at a time.

RQ1a: Does minimal-data elicitation, demonstrated at 32B parameters in English, transfer to the 0.6B to 4B range in Bengali?

RQ1b: If elicitation does not improve accuracy, does it change model behaviour at all, and in which direction?

TigerLLM-1B and TituLLM-1B match in size but differ in pretraining depth. TituLLM-1B and TituLLM-3B share a pretraining approach but differ in size. The Qwen3 ladder isolates scale with Bengali exposure held near zero. This design lets us attribute any difference to one factor at a time, which no prior work in this space has done.

II. LITERATURE REVIEW

A. The Low-Resource Reasoning Baseline

Reasoning in low-resource languages is a known weak point, and Bengali is no exception. Al Nazi et al. (2025) tested six LLMs on four Bengali tasks under zero-shot, few-shot and chain-of-thought prompting. The best result, GPT-4o with three-shot CoT, reached 99% F1 on sentiment, but CoT was not reliably helpful and in several cases reduced accuracy through hallucination. The failure mode is therefore not simply insufficient capacity. Reasoning behaviour itself is unstable in this setting.

BenNumEval (Ahmed et al., 2025) sharpens the point. Across 3.2k samples in six numerical reasoning categories, DeepSeek-R1-Distill-Llama-70B, built from top-tier English reasoning distillation, collapsed to 4.7% accuracy in native Bengali. That is far below Gemini-2.0-flash’s 79.6%, and below several non-distilled baselines. This is not merely a failure to transfer; it suggests distilling English reasoning traces can actively damage existing multilingual ability. SO-MADHAN (Paul et al., 2025) adds an important qualifier. Using 8.8k expert-translated Bengali word problems, Llama-3.3-70B reached 88%, but smaller models largely failed to adapt to native-language reasoning at all. The elicitation problem therefore interacts with model size, which is precisely what this paper sets out to measure.

B. The Backbone Debate: Scale versus Targeted Depth

This is the central disagreement in the literature, and it splits three ways rather than two.

The **targeted-depth** position is TigerLLM (Raihan & Zampieri, 2025). Pretrained on only about 10M tokens of Bengali textbook content plus 100k instruction pairs distilled from GPT-4o and Claude, its 1B variant reportedly outperformed 11B generalist models, reaching 68% on BanglaRQA. The claim is not that small models can be competitive. It is that targeted depth substitutes for scale outright.

The **scale** position pushes back. Yong et al. (2025) found that test-time scaling becomes reliable only at 3B parameters or above, with a 14B model at a sweet spot, and noted that Bengali’s tokenizer makes reasoning roughly $3.5\times$ more expensive than English. MathMist (Sobhani et al., 2026), a 30k-pair benchmark across 13 languages, reports that scaling Qwen from 0.6B to 14B shrank the Bengali–English accuracy gap from 7% to 1%, with smaller models showing what the authors call linguistic brittleness during CoT.

Classical scaling work supplies the mechanism. Kaplan et al. (2020) showed that language modelling loss depends strongly on parameter count and only weakly on architectural shape, to the point where aspect ratio can vary by a factor

of 40 with little effect. Wei et al. (2022) is more directly relevant to our range. They report that three-digit arithmetic emerges only at roughly 13B parameters, and that chain-of-thought prompting surpasses standard prompting on GSM8K only at around 100B parameters. If arithmetic competence and CoT benefit both emerge above 13B, then the 0.6B to 4B range studied here sits entirely below the threshold at which the relevant abilities have been shown to appear.

A third position complicates both. TituLLM (Nahin et al., 2025) pretrained on roughly 37B tokens of general-domain Bengali with an extended tokenizer, and found that this helped commonsense-style tasks but not broad-knowledge tasks such as MMLU. The authors partly attribute this to single-epoch training and the absence of instruction tuning. On this account pretraining depth helps some capabilities and not others.

The problem common to all three is that none of them isolates scale from pretraining depth. TigerLLM’s depth claim exists only at 1B and 9B. TituLLM’s scale claim exists only inside its own general-corpus recipe. The two variables are always entangled. Closing that gap, by fixing scale at 1B while varying depth and fixing the recipe while varying scale, is the design of this paper.

C. The Less-Is-More Elicitation Paradigm

A second strand argues that surfacing reasoning in an already-capable model takes remarkably little data. This is what makes a small Bengali training set a reasonable intervention rather than an underpowered shortcut.

LIMA (Zhou et al., 2023) states the underlying claim most directly, as the Superficial Alignment Hypothesis: a model’s knowledge is acquired almost entirely during pretraining, and alignment principally teaches it which format and style to use when responding. LIMO (Ye et al., 2025) carries the argument into mathematics, reporting that 800 curated reasoning traces elicited more sophisticated reasoning than training on the 100k-sample NuminaMath dataset, with gains saturating around 1.6k samples or 32B parameters. s1 (Muennighoff et al., 2025) arrives somewhere similar by a different route: 1k traces plus budget forcing let s1-32B beat o1-preview on AIME24 by 27%. LESS (Xia et al., 2024) offers a mechanism, showing that selecting 5% of a 270k-point pool by projected low-rank gradient relevance beat training on the full set. Functional relevance beats raw volume.

Two features of this literature matter here. First, every demonstration is at 32B parameters or above. Second, all of them are in high-resource languages. LIMO’s own stated precondition, that the base model must already hold dense and relevant knowledge from pretraining, is an empirical claim about the backbone. Whether it holds at 0.6B to 4B in Bengali is exactly what we test.

D. The Elicitation Tax: Native Reasoning versus English Pivot

There is a measurable cost to reasoning in Bengali specifically. Qi et al. (2025), using the 370-question, 11-language XReasoning benchmark, found that forcing native Bengali

reasoning instead of pivoting through English dropped accuracy on hard tasks such as AIME from 26% to 17%. Their proposed fix was tested on at most 817 samples, so it is a partial answer at best. Yong et al. (2025) adds that Bengali’s tokenizer makes reasoning roughly $3.5\times$ more expensive, and that even where in-domain scaling works, generalisation to out-of-domain cultural knowledge stays weak. Taken together with the BenNumEval collapse, these results suggest that interventions meant to improve Bengali reasoning may trade language adherence against accuracy. We observe that tension directly in our own results.

E. Adaptation Method and Its Limits

Because this study fine-tunes with LoRA rather than full parameter updates, the method itself is a variable that has to be accounted for.

LoRA (Hu et al., 2022) freezes pretrained weights and injects trainable low-rank matrices. It was introduced with results showing parity or better against full fine-tuning on GLUE-style tasks, for instance 91.3 versus 91.1 average GLUE for DeBERTa-XXL using 4.7M of 1.5B parameters. Those tasks are largely classification and generation within a model’s existing competence, though. Biderman et al. (2024) tests the harder case directly. On mathematics, measured as GSM8K strict match after continued pretraining, full fine-tuning reached 0.293 while LoRA at rank 256 reached 0.202 and rank 16 only 0.158. On code instruction tuning, full fine-tuning reached 0.470 against 0.358 for rank 16. Their conclusion is that “in the standard low-rank settings, LoRA substantially underperforms full finetuning,” which they attribute to full fine-tuning finding perturbations of rank 10 to 100 times greater than typical LoRA configurations. They also report the converse: LoRA forgets less, keeping the tuned model’s behaviour closer to the base model.

This has two consequences here. It predicts that a rank-16 configuration will under-acquire new mathematical capability, which bears directly on how our null result should be read. It also predicts that base behaviour will be comparatively preserved, which is consistent with what we observe. We note that Biderman et al. train only at 7B, outside the range studied here.

Domain-adaptive pretraining supplies the other half of the frame. Both TigerLLM and TituLLM are instances of continued pretraining on a target domain, differing by more than three orders of magnitude in token budget, and neither reports an ablation isolating that budget from the base model it starts from.

F. The Cost of Bengali Tokenization

Tokenizer efficiency is usually treated as an implementation detail. For Bengali it is a first-order constraint. Ahia et al. (2023) show that languages with their own scripts require up to $5\times$ more tokens than English to convey identical content, that API costs rise correspondingly, and that high fragmentation can stop even a single in-context example from fitting in a fixed context window. Petrov et al. (2023) quantify this per

model, reporting Bengali tokenization premiums that range from 9.65 for GPT-2 and 5.84 for ChatGPT down to 1.01 for the Indic-focused MuRIL. That is close to a tenfold spread in how much Bengali text fits in the same budget.

The backbones studied here span that spread, though by accident rather than design. Llama 3 (Llama Team, AI@Meta, 2024) adopts a 128k-token vocabulary explicitly to improve non-English compression, and TituLLM extends it further for Bengali. Gemma 3 (Gemma Team, Google DeepMind, 2025) ships a 262k-entry SentencePiece vocabulary described as “more balanced for non-English languages,” which TigerLLM inherits without doing any tokenizer work of its own. Qwen3 (Qwen Team, 2025) uses a 151,669-entry BBPE vocabulary covering 119 languages. These choices were made for reasons unrelated to Bengali, yet they determine how much Bengali reasoning fits inside a fixed training budget, and therefore which models can be trained at all on consumer hardware.

G. Gap Statement

Three gaps remain open for RQ1.

First, no Bengali-language study separates parameter scale from pretraining depth as independent factors. The works in Section 2.2 always vary size and corpus design together, so existing work cannot tell us which one is responsible.

Second, minimal-data elicitation has been demonstrated only at 32B parameters or above, and only in high-resource languages, while the abilities it aims to surface, arithmetic competence and chain-of-thought benefit, have themselves been reported to emerge only above roughly 13B (Wei et al., 2022). Whether the paradigm holds below that threshold, in a language carrying a $3.5\times$ tokenization penalty, is untested.

Third, the interaction between adaptation method and this question is unexamined. LoRA is the only practically available option at consumer scale, yet it is documented to under-acquire new capability relative to full fine-tuning (Biderman et al., 2024), and no prior work reports what minimal-data elicitation does under that constraint in a low-resource language.

III. METHODOLOGY

A. Experimental Design

The study is a controlled experiment with two independent variables, parameter scale and targeted Bengali pretraining depth, and one intervention, LIMO-style minimal-data fine-tuning. Every model is evaluated twice on the same held-out problems, once as released and once after fine-tuning, so each model acts as its own control.

Seven backbones were selected so that the two independent variables move separately. Table I summarises them. TigerLLM-1B and TituLLM-1B are close in size but differ by roughly $3,700\times$ in targeted Bengali pretraining. TituLLM-1B and TituLLM-3B share a pretraining recipe and differ only in size. The Qwen3 family spans 0.6B to 4B with no Bengali-specific pretraining, isolating scale.

TigerLLM is instruction-tuned and TituLLM’s base model is not, a comparison between them would confound pre-training depth with instruction tuning. We therefore evaluate

the instruction-tuned TituLLM variants as the primary depth comparison and retain the base variant as an instruction-tuning ablation.

TABLE I
THE SEVEN BACKBONES UNDER TEST. BENGALI DEPTH RECORDS WHETHER THE MODEL RECEIVED BENGALI-SPECIFIC CONTINUED PRETRAINING BEYOND A GENERAL MULTILINGUAL MIX. PARAMETER COUNTS ARE TOTAL, NOT TRAINABLE; LoRA TRAINS ROUGHLY 0.5% OF THEM.

Model	Base	Bengali PT	Params
TigerLLM-1B-it	Gemma3	~10M tokens	1.0B
TituLLM-1B v2.0	Llama-3.2	~37B tokens	1.2B
TituLLM-1B-Instruct	Llama-3.2	~37B tokens	1.2B
TituLLM-3B-Instruct	Llama-3.2	~37B tokens	3.2B
Qwen3-0.6B	Qwen3	incidental	0.6B
Qwen3-1.7B	Qwen3	incidental	1.7B
Qwen3-4B	Qwen3	incidental	4.0B

B. Training Data Curation

Training data was drawn from the SFT split of the Ganit dataset (Roy Dipta et al., 2026), a Bengali mathematical reasoning corpus of 16,868 problems, each carrying a full Bengali reasoning trace and a difficulty score. The difficulty score is the number of times Qwen3-32B solved the problem across 32 sampled attempts, so it records difficulty relative to a 32B model. The pool is assembled from four upstream sources: numina-math-cot-bn (13,160 problems), SOMAD-HAN (3,487), mCoT-MATH-bn (193) and s1k-32-Bangla (28). Its own difficulty tags skew easy (10,065 easy, 6,411 olympiad, 258 hard, 134 medium), but the solve-rate column tells a harsher story: 5,645 problems, a third of the corpus, were never solved by Qwen3-32B in any of 32 attempts, and 91.7% of that unsolvable band comes from numina-math-cot-bn alone. We treat the zero band as unreliable rather than merely hard, and exclude it.

Figure 1 shows the pipeline. Decontamination ran first, before any difficulty filtering, against 4,080 problems pooled from three public Bengali mathematics benchmarks: Bn-MGSM (250 problems), Bn-MSVAMP (1,000) and BenNumEval (2,830 across five task types). A training problem was removed if it matched a benchmark problem on either of two signals. The first is a numeric fingerprint, the hash of the sorted set of numbers appearing in the normalised problem text. Numbers survive re-translation even when every surrounding word changes, so this catches exactly the case that n-gram or MinHash deduplication misses. The second is token Jaccard overlap at a threshold of 0.7, which catches near-identical wording. Normalisation maps Bengali numerals to Western digits, lowercases, strips punctuation and collapses whitespace, so the two numeral conventions present in the corpus compare on equal footing. Together these removed 3,315 problems and left 13,553.

Following LIMO’s selection criterion, we then retained problems in the window $1 \leq \text{correct_counts} \leq 16$, which reduced 13,553 to 873. A final near-duplicate pass within the surviving set removed nothing, which is itself mild evidence that the pool is not padded with reworded repeats. The

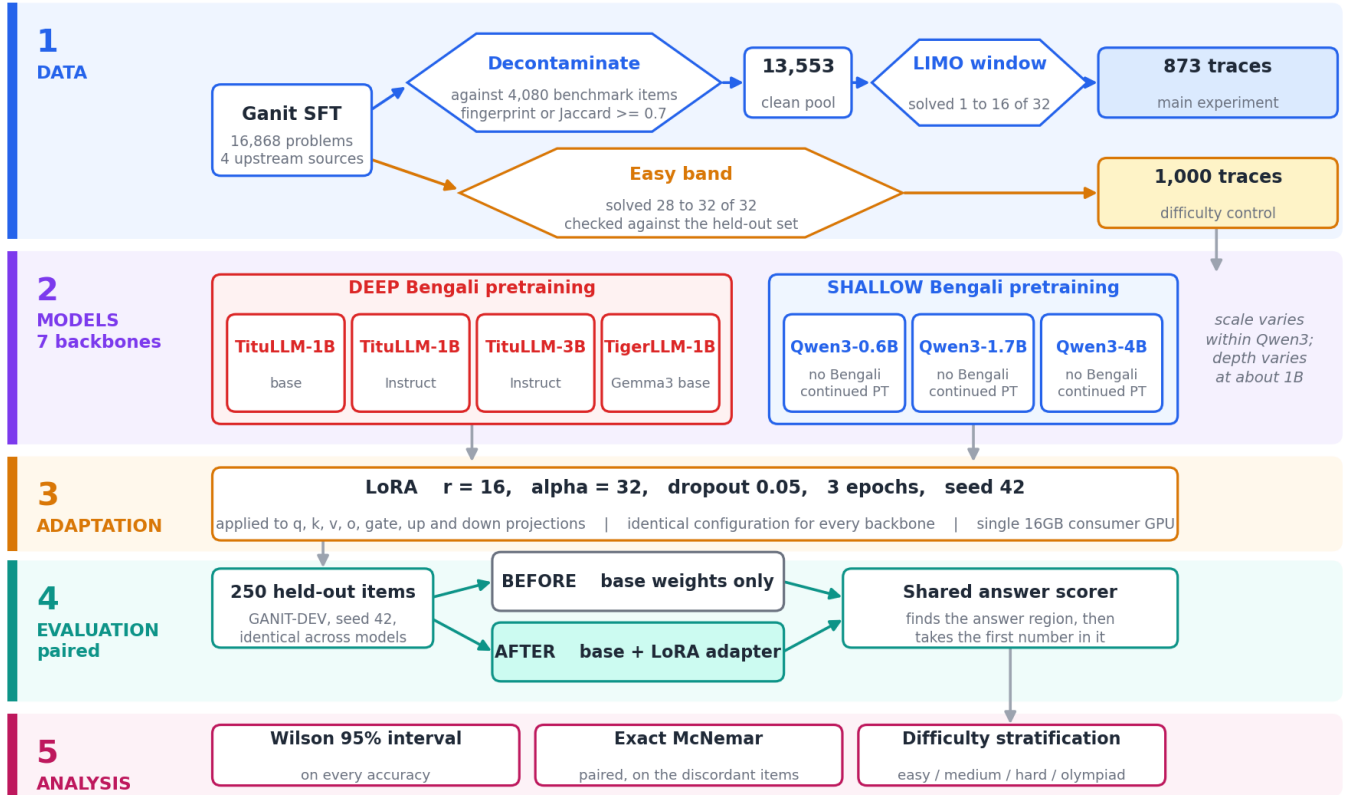


Fig. 1. The study in full. Decontamination against three public Bengali mathematics benchmarks precedes difficulty filtering, so the LIMO window is applied to an already-clean pool. The two training sets differ only in the difficulty band they are drawn from. Every backbone receives the identical LoRA configuration and is evaluated on the identical held-out items in both conditions, which makes every comparison in this paper a paired one.

resulting set is heavily weighted towards hard problems: 74.6% carry an olympiad difficulty tag and 25.3% are tagged hard, with a mean pass rate of 3.3 out of 32.

A second training set was constructed later to test whether that difficulty distribution explains our results. Holding every other choice constant, we selected 1,000 traces from the band $\text{correct_counts} \geq 28$, giving a mean pass rate of 31.8 out of 32 and a difficulty composition of 100% easy. Section IV-E reports that experiment. This set was decontaminated directly against the held-out evaluation items rather than against the public benchmarks, using the same numeric fingerprint idea. That filter is deliberately aggressive and removed 1,782 candidates.

The evaluation set was independently checked against the 873 training traces by identifier match, exact normalised text match, and numeric fingerprint combined with Jaccard overlap. One item matched and was dropped. Neither training set shares an identifier or problem text with the evaluation set, and the two training sets are disjoint from each other because they are drawn from non-overlapping difficulty bands. One limitation is worth stating plainly: Ganit carries no upstream English source identifiers, so fingerprint and Jaccard matching are proxies, and a sufficiently free paraphrase could survive both filters.

C. Evaluation Set

The held-out set is 500 problems sampled from GANIT-DEV with seed 42, stratified by difficulty: 145 easy, 127 medium, 129 hard and 99 olympiad. Gold answers are short final values rather than worked solutions.

Compute constraints on a single consumer GPU limited scoring to the first 250 items of that shuffled set for every model, and to the first 100 for TituLLM-1B base, which was withdrawn from the extension after producing degenerate output at 0 to 2% accuracy across its first hundred items. Every number in this paper is therefore reported on 250 items unless stated otherwise. Because the set was shuffled before truncation, the scored subset preserves the difficulty mix: 69 easy, 69 medium, 67 hard and 45 olympiad. All models see the same items in the same order, so every before-and-after comparison in this paper is paired.

D. Fine-Tuning Protocol

All models were fine-tuned with LoRA (Hu et al., 2022) using an identical configuration: rank 16, $\alpha = 32$, dropout 0.05, applied to the attention and MLP projection matrices (q_proj , k_proj , v_proj , o_proj , $gate_proj$, up_proj , $down_proj$). Training used a learning rate of 2×10^{-4} with cosine decay and 3% warmup, three epochs, per-device batch size 1 with gradient accumulation 16 for an

effective batch of 16, bfloat16 precision, gradient checkpointing, and seed 42. Loss was computed on the target tokens only, with prompt tokens masked.

The training target is the full Bengali reasoning trace from the `Ganit.messages` field, not the short final answer. Training on the bare answer would teach the model to emit a value and stop, which is close to the opposite of reasoning elicitation.

The Bengali instruction text is byte-identical for every model. Only the wrapper around it differs: completion-style models receive the instruction followed by a Bengali solution cue (*somadhan*, meaning “solution”), while instruction-tuned models receive it through their own chat template. This choice was made empirically rather than assumed. We probed both formats on each model and selected whichever produced a lower token-cap rate, on the reasoning that a format a model was never trained in prevents it from terminating. For Qwen3 models, thinking mode was disabled so that every backbone receives the same intervention.

All experiments ran on a single NVIDIA RTX 5060 Ti with 16GB of VRAM. Total training time across all adapters was approximately 4.4 GPU-hours.

E. Availability and Computational Environment

All experiments ran under Python 3.11.16 with PyTorch 2.8.0+cu128, Transformers 4.57.6 and PEFT 0.20.0, on a single NVIDIA RTX 5060 Ti with 16GB of VRAM. Seed 42 was set for Python, NumPy and PyTorch in every run, and all generation is greedy, so no sampling randomness enters the results. Deterministic CUDA algorithms were not enabled, because the throughput cost was prohibitive on this hardware; the consequence is that training losses reproduce to roughly three decimal places rather than exactly, while greedy generations are stable.

The curation notebook, the training and evaluation harness, the shared answer scorer with its executable test suite, and the analysis scripts that regenerate every table, figure and reported number in this paper from the stored model outputs are available at GitHub. The exact environment record, including package versions and the reproducibility caveats above, is committed as `environment.json`.

F. Evaluation Protocol

Generation is greedy, with a repetition penalty of 1.1 and a per-model generation budget set to the 99th percentile of that model’s own training-target length. This last point matters and is not a detail. An equal token budget is not an equal opportunity when tokenizers differ by a factor of three. The same 873 Bengali traces require a median of 397 tokens under TituLLM’s tokenizer, 428 under TigerLLM’s, and 1,245 under Qwen3’s. A uniform 1,024-token cap would leave the Bengali models room to spare while truncating 42% of what Qwen3 was trained to produce. Qwen3 models therefore receive 2,048 tokens and the Bengali models 1,024, with the budget recorded for every prediction.

Generation halts on any end-of-sequence marker the tokenizer knows, rather than only its declared `eos_token`. This

is necessary because TituLLM declares `<|eot_id|>`, a chat turn terminator, while being prompted in completion format where that token never follows free text. Under a single-token stopping rule the model has no reachable way to stop.

G. Answer Extraction and Scoring

Answers are scored by exact match after normalisation. Extraction proceeds by narrowing to an answer region, using `<answer>` tags, then `\boxed{}`, then the Bengali answer marker *uttar* (“answer”), and finally the last numeral in the generation, before taking the first numeric token inside that region. Normalisation maps Bengali numerals to ASCII, strips currency, percentage and LaTeX markup, and compares numerically where possible.

Taking the first number inside the answer region rather than the whole region is essential. An earlier version returned the entire remainder of the line after the answer marker. A correct answer written naturally, for example *uttar: 20 jon chhatro* (“answer: 20 students”), was compared in full against a gold value of 20, failed to normalise, and was scored wrong even though the model was right. The scorer is distributed with an executable test suite built from verbatim model outputs rather than synthetic cases.

Generations that produce no parseable answer are recorded as format failures and reported separately from wrong answers.

H. Statistical Analysis

Because every model is evaluated on the same items, before-and-after comparisons are paired and use the exact McNemar test on discordant pairs. Comparisons between different models use Fisher’s exact test. Proportions are reported with Wilson score intervals, which behave correctly near zero where the normal approximation does not. Continuous measures such as generation length and Bengali character ratio are reported as mean paired differences with 95% confidence intervals, and treated as significant when the interval excludes zero.

Every before-and-after comparison is reported with a paired risk difference Δ and its 95% confidence interval alongside the p -value, because a p -value alone says whether an effect is distinguishable from zero but not how large it could plausibly be. Where the two disagree we treat the exact test as authoritative: the Wald interval on a paired difference is anti-conservative when the discordant counts are very small, which happens here for TituLLM-1B-Instruct with eight discordant pairs.

We report significance tests rather than raw differences throughout because accuracies in this study are low. At 3% accuracy on 250 items, a difference of three correct answers looks like a trend and is not one.

No correction for multiple comparisons is applied. Seven McNemar tests, three Fisher tests and seven sign tests are reported. Since every McNemar result is null, a correction could only make them more null and would not change any conclusion; the Fisher and sign-test results survive Bonferroni correction at $\alpha = 0.05$ by several orders of magnitude. Readers should nonetheless treat the single borderline value, $p = 0.070$ for TituLLM-1B-Instruct, as weaker than it looks in isolation.

IV. RESULTS

A. Scale Predicts Bengali Reasoning; Pretraining Depth Does Not

Figure 2 is the central result. Plotting untuned accuracy against parameter count separates the models cleanly into two regimes, and the separating variable is not the one the field’s Bengali-specific model releases would predict.

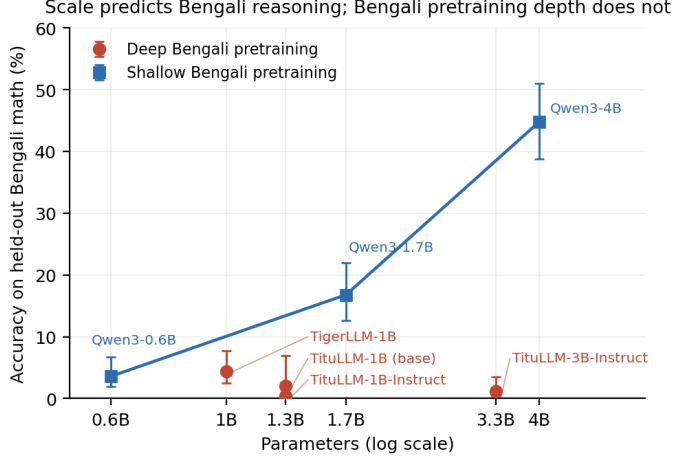


Fig. 2. Exact-match accuracy on held-out Bengali mathematics word problems, plotted against total parameter count, for all seven backbones as released and before any fine-tuning. Each model is scored on the same 250 items (100 for TituLLM-1B base); vertical bars are Wilson 95% intervals and the horizontal axis is logarithmic. The line joins members of the Qwen3 family and is a visual guide, not a fitted trend. Notice that the four models with deep Bengali pretraining sit flat near the floor across a threefold range of parameters, while the three with only incidental Bengali coverage climb steeply with scale.

Within the Qwen3 family, which received no Bengali-specific pretraining beyond whatever appears in a general multilingual mix, accuracy rises monotonically and steeply: 3.6% at 0.6B, 16.8% at 1.7B, and 44.8% at 4B. Every adjacent step is significant under an unpaired Fisher exact test: $p = 1.0 \times 10^{-6}$ from 0.6B to 1.7B, $p = 1.3 \times 10^{-11}$ from 1.7B to 4B, and $p = 7.4 \times 10^{-30}$ across the full ladder. That is better than an order of magnitude of improvement across less than one order of magnitude of parameters.

The Bengali-specialised models show no such gradient. TituLLM-1B-Instruct reaches 0.4%, TituLLM-1B base 2.0%, TigerLLM-1B 4.4%, and TituLLM-3B-Instruct 1.2%. Tripling the parameter count from 1B to 3.2B buys nothing. All four sit inside each other’s confidence intervals and all four sit near the floor.

The sharpest single comparison is TituLLM-3B-Instruct against Qwen3-1.7B. TituLLM-3B-Instruct has roughly twice the parameters and was continually pretrained on a large Bengali corpus with a tokenizer extended for the language. Qwen3-1.7B has neither advantage. Qwen3-1.7B scores 16.8% against TituLLM-3B-Instruct’s 1.2% (Fisher exact, $p = 1.7 \times 10^{-10}$). The advantage is not an artifact of one difficulty tier. The smaller, Bengali-shallow model wins at easy (17.4%

against 1.4%), medium (20.3% against 1.4%), hard (13.4% against 1.5%) and olympiad (15.6% against 0.0%) alike.

This answers RQ1 directly, and answers it in the negative. Targeted Bengali pretraining depth, as delivered by the two publicly released Bengali model families we could obtain, does not substitute for scale on mathematical reasoning. Whatever those models gained from Bengali continued pretraining, and they did gain something, since their tokenizers are far more efficient on Bengali text, it did not transfer into mathematical competence.

B. Minimal-Data Elicitation Moves No Model

Table II and Figure 3 report the paired before-and-after comparison for all seven backbones.

We report accuracy rather than precision, recall or F1 because the task is free-form generation of a numeric answer scored by exact match, not classification: there are no classes over which those metrics are defined and no confusion matrix to construct. In their place we report the diagnostics that do carry information for generative scoring, namely format-failure rate, token-cap rate, language adherence, per-difficulty accuracy (Table III) and answer retention (Table IV).

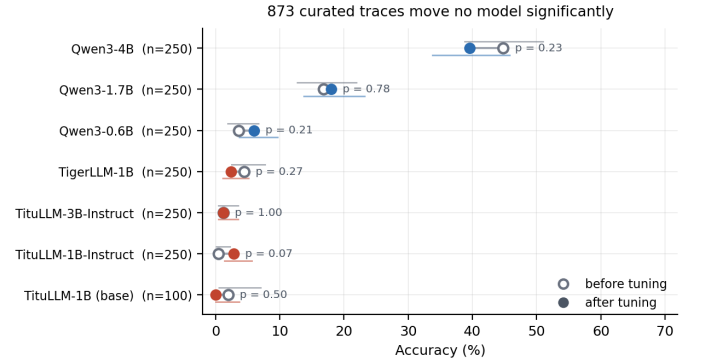


Fig. 3. Exact-match accuracy for each backbone before and after LoRA fine-tuning on the same 873 curated Bengali traces, scored on the same held-out items in both conditions ($n = 250$; $n = 100$ for TituLLM-1B base). Hollow markers are the untuned model, filled markers the adapter, and the thin lines behind each marker are Wilson 95% intervals. p values are two-sided exact McNemar tests on the discordant pairs. Notice that the intervals for the two conditions overlap substantially for every model, and that no comparison crosses the 0.05 threshold.

Not one of the seven comparisons reaches significance. The smallest p value in the table is 0.070, for TituLLM-1B-Instruct, on a change from 0.4% to 2.8% that amounts to six additional correct answers out of 250. Three models moved up, three moved down, and one did not move at all. That distribution is what noise looks like.

The result is worth stating carefully, because a null result invites two different readings and only one of them is supported here. The reading we can defend is that 873 curated traces do not elicit latent mathematical reasoning in Bengali at the 0.6B to 4B scale. The reading we cannot defend is that LIMO-style elicitation does not work, full stop. Our largest model is 4B. Wei et al. (2022) place the threshold for arithmetic emergence near 13B and for chain-of-thought benefit substantially higher,

TABLE II

ACCURACY BEFORE AND AFTER LIMO-STYLE FINE-TUNING ON 873 CURATED BENGALI REASONING TRACES, ON THE SAME 250 HELD-OUT ITEMS PER MODEL (TITULLM-1B BASE ON 100). INTERVALS ON EACH ACCURACY ARE WILSON 95%. Δ IS THE PAIRED RISK DIFFERENCE IN PERCENTAGE POINTS WITH ITS 95% CONFIDENCE INTERVAL; IT IS THE EFFECT SIZE, AND IT IS REPORTED BECAUSE A p -VALUE ALONE DOES NOT BOUND HOW LARGE THE CHANGE COULD BE. p IS A TWO-SIDED EXACT MCNEMAR TEST ON THE DISCORDANT PAIRS. NOTICE THAT NO COMPARISON REACHES SIGNIFICANCE AND THAT SIX OF THE SEVEN Δ INTERVALS CONTAIN ZERO. THE EXCEPTION IS TITULLM-1B-INSTRUCT, WHOSE INTERVAL IS COMPUTED FROM ONLY EIGHT DISCORDANT PAIRS; THE WALD INTERVAL IS ANTI-CONSERVATIVE AT THAT COUNT AND THE EXACT TEST ($p = 0.070$) IS THE RELIABLE READING.

Model	Params	Bn depth	n	Before	After	Δ (pp)	95% CI	p
TituLLM-1B (base)	1.0B	deep	100	2.0%	0.0%	-2.0	[-4.7, +0.7]	0.500
TituLLM-1B-Instruct	1.0B	deep	250	0.4%	2.8%	+2.4	[+0.2, +4.6]	0.070
TigerLLM-1B	1.0B	deep	250	4.4%	2.4%	-2.0	[-4.8, +0.8]	0.267
TituLLM-3B-Instruct	3.2B	deep	250	1.2%	1.2%	+0.0	[-1.9, +1.9]	1.000
Qwen3-0.6B	0.6B	shallow	250	3.6%	6.0%	+2.4	[-0.7, +5.5]	0.210
Qwen3-1.7B	1.7B	shallow	250	16.8%	18.0%	+1.2	[-4.4, +6.8]	0.780
Qwen3-4B	4.0B	shallow	250	44.8%	39.6%	-5.2	[-13.0, +2.6]	0.228

TABLE III

ACCURACY BY DIFFICULTY TIER, BEFORE AND AFTER FINE-TUNING, FOR EVERY BACKBONE ON THE SAME HELD-OUT ITEMS. TIER COUNTS ARE 69 EASY, 69 MEDIUM, 67 HARD AND 45 OLYMPIAD FOR THE MODELS EVALUATED ON 250 ITEMS, AND 30/24/29/17 FOR TITULLM-1B BASE ON 100. NOTICE THAT THE QWEN3 ADVANTAGE HOLDS IN EVERY TIER RATHER THAN COMING FROM ONE BAND, AND THAT NO TIER SHOWS A CONSISTENT POST-TUNING GAIN.

Model	Easy	Medium	Hard	Olympiad
TituLLM-1B (base)	0.0 $\rightarrow$ 0.0	4.2 $\rightarrow$ 0.0	3.4 $\rightarrow$ 0.0	0.0 $\rightarrow$ 0.0
TituLLM-1B-Instruct	1.4 $\rightarrow$ 5.8	0.0 $\rightarrow$ 2.9	0.0 $\rightarrow$ 1.5	0.0 $\rightarrow$ 0.0
TigerLLM-1B	7.2 $\rightarrow$ 2.9	4.3 $\rightarrow$ 2.9	1.5 $\rightarrow$ 3.0	4.4 $\rightarrow$ 0.0
TituLLM-3B-Instruct	1.4 $\rightarrow$ 1.4	1.4 $\rightarrow$ 1.4	1.5 $\rightarrow$ 1.5	0.0 $\rightarrow$ 0.0
Qwen3-0.6B	2.9 $\rightarrow$ 8.7	4.3 $\rightarrow$ 4.3	4.5 $\rightarrow$ 9.0	2.2 $\rightarrow$ 0.0
Qwen3-1.7B	17.4 $\rightarrow$ 21.7	20.3 $\rightarrow$ 20.3	13.4 $\rightarrow$ 19.4	15.6 $\rightarrow$ 6.7
Qwen3-4B	59.4 $\rightarrow$ 55.1	44.9 $\rightarrow$ 34.8	40.3 $\rightarrow$ 40.3	28.9 $\rightarrow$ 22.2

and every model in this study sits below both. A negative result collected entirely below a capability threshold is evidence about that region, not about the method.

Qwen3-4B is the one row where the point estimate moved by a visible margin, from 44.8% down to 39.6%. The direction survived the extension from 100 items to 250, but the significance did not arrive with it: 56 items were lost against 43 gained, giving $\Delta = -5.2$ percentage points with a 95% confidence interval of -13.0 to $+2.6$, and an exact McNemar $p = 0.228$. The interval contains zero and is wide enough to accommodate a modest gain, so we do not read a real degradation into it. It is worth noting that this is the only model with enough baseline accuracy for a degradation to have been visible at all.

C. Output Behaviour Changed Even Though Accuracy Did Not

The tuning was not inert. It changed what the models produced; it simply did not change whether the produced answers were right.

The clearest case is TituLLM-1B base. Untuned, it hit the generation cap on 100% of items, meaning it never once emitted a stop token and every completion was truncated at the token budget. After fine-tuning on 873 traces, that fell to 47%. The model learned, from a very small number of examples, what a finished Bengali reasoning trace looks like and when to stop producing one. Its accuracy over the same items went from 2.0% to 0.0%.

Across the other backbones the cap rate mostly moved the other way. It rose after tuning for four of the seven models, most sharply for Qwen3-0.6B (4.8% to 36.8%) and TituLLM-3B-Instruct (8.0% to 29.2%). Trained on long worked solutions, those models produced longer outputs. They did not produce more correct ones. Qwen3-4B is the informative exception: the one model with real baseline competence barely moved on this measure at all, from 8.8% to 7.6%, which is what we would expect from a model that already knew how to finish a worked solution before it saw any of ours.

Stopping behaviour, then, moved in whichever direction the base model had the most room to move in, and in no case did it move accuracy with it. The next section takes this further and asks what the adapter changed at the level of individual items.

D. What the Fine-Tuning Actually Did

A null result invites the objection that nothing happened at all: that the adapter never took, or that LoRA at rank 16 is too small to move a model. The item-level data rules that out. A great deal happened. It simply was not learning to reason.

Table IV reports answer retention, meaning the share of items a model answered correctly before fine-tuning that it still answered correctly afterwards. If 873 traces were teaching mathematical reasoning, retention should be high and the gains should sit on top of it. That is not the pattern.

Qwen3-4B answered 112 items correctly before tuning and kept 56 of them. It also answered 43 items correctly that it had

TABLE IV

ANSWER RETENTION. OF THE ITEMS EACH MODEL ANSWERED CORRECTLY BEFORE FINE-TUNING, HOW MANY IT STILL ANSWERED CORRECTLY AFTERWARDS, AND HOW MANY ITEMS CHANGED OUTCOME IN EITHER DIRECTION. NOTICE THAT NO MODEL RETAINED EVEN HALF OF WHAT IT ALREADY KNEW, WHICH IS WHY THE SMALL NET CHANGES IN TABLE II UNDERSTATE HOW MUCH THE ADAPTER MOVED.

Model	Correct before	Kept	Retention	Items flipped
Qwen3-4B	112	56	50%	99 / 250
Qwen3-1.7B	42	18	43%	51 / 250
Qwen3-0.6B	9	4	44%	16 / 250
TigerLLM-1B	11	2	18%	13 / 250
TituLLM-3B-Instruct	3	0	0%	6 / 250
TituLLM-1B-Instruct	1	0	0%	8 / 250

failed before. In total 99 of 250 items changed their outcome, close to 40% of the evaluation set, while net accuracy moved by 13 items. The same shape holds down the scale: no model retained even half of what it already knew. The adapter did not add a layer of competence on top of the base model. It substantially rewrote which problems the model could solve, and the replacement set happened to be slightly worse than the original.

The mechanism is visible in the generated text. Before fine-tuning, Qwen3-4B closed 225 of 250 completions with the Bengali word for “answer” followed by its result, a convention it adopted on its own, since the evaluation prompt asks only that the model answer in Bengali and specifies no format. After fine-tuning it did so on 3 of 250. In its place the model adopted the format of the training corpus: Ganit’s traces are written as a `<think>` block followed by an `<answer>` block, and after tuning every one of the 250 completions opened a `<think>` block. Median output length fell from 1,046 tokens to 887, the proportion of Bengali characters fell from 87.0% to 82.0%, and the number of explicit step markers roughly halved.

This is the Superficial Alignment Hypothesis (Zhou et al., 2023) observed from the unfavourable side. In LIMA’s setting a small number of examples exposes capability the pretrained weights already hold, and the result is a useful model. Here the same small number of examples reshaped format, length, language mix and stopping behaviour just as comprehensively, and exposed nothing, because at this scale there was little underneath to expose. Style transferred completely. Reasoning did not transfer at all.

A sensitivity check on the format change: Because the output format changed, it is worth confirming that the accuracy drop is not an artifact of answer extraction. Our scorer looks for an `<answer>` block first, then a boxed expression, then the Bengali answer marker, and only if none of those are present does it fall back to taking the last number in the completion. After fine-tuning, 230 of 250 Qwen3-4B completions contained a well-formed `<answer>` block and were therefore parsed by the strongest branch of the extractor. The remaining 20 exhausted the generation budget inside the `<think>` block and were scored by the weak fallback. Of the 56 items Qwen3-4B lost, 45 produced a well-formed `<answer>` that was simply wrong. The failures are arithmetic, not formatting.

On one item with gold answer 16 the tuned model emitted `<answer>0</answer>`; on another with gold 126 it emitted `<answer>14</answer>`. In both cases the extractor read exactly what the model wrote.

To bound the effect, we recomputed accuracy crediting every fallback-scored item as correct, which is the most generous assumption available. Post-tuning accuracy under that assumption is 46.8%, against a measured 39.6% and a pre-tuning 44.8%. The true value therefore lies somewhere between a loss of 5.2 points and a gain of 2.0 points. We cannot sign the effect, which is consistent with the exact McNemar test returning $p = 0.228$, and the substantive conclusion is unchanged in either direction: 873 curated traces did not produce a detectable improvement in Bengali mathematical reasoning at this scale.

E. The Null Is Not a Difficulty-Calibration Artifact

The most obvious objection to Section IV-B is that our training set was simply too hard. The LIMO window selects problems a 32B model solves between 1 and 16 times out of 32, which for a 1.7B model may be far outside the range it can learn anything from. If so, the null would say more about our data selection than about elicitation.

Figure 4 reports that test. Holding the backbone, the LoRA configuration, the seed, the prompt and the evaluation items constant, we built a second training set of 1,000 traces from the opposite end of the difficulty distribution, requiring at least 28 correct out of 32, and trained the same Qwen3-1.7B on it.

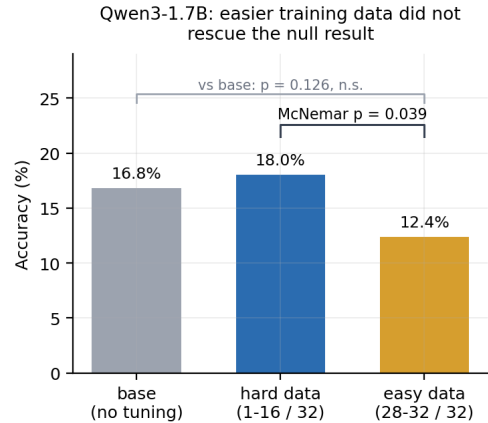


Fig. 4. Qwen3-1.7B under three conditions, scored on the same 250 held-out items, with backbone, LoRA configuration, seed, prompt and evaluation items held fixed so that training-data difficulty is the only thing that varies. The easy-data adapter is significantly worse than the hard-data adapter (exact McNemar, $p = 0.039$); against the untuned base it is lower but not significantly so ($p = 0.126$), as marked. Notice that easier, better-calibrated training data did not rescue the null result and if anything moved accuracy the wrong way.

The easy-data adapter reached 12.4%, against 18.0% for the hard-data adapter and 16.8% for the untuned base. The gap between the two adapters is significant (exact McNemar, 27 items lost against 13 gained, $p = 0.039$), and it runs in the opposite direction to the objection. Easier, better-calibrated training data did not rescue the null. It made results worse.

Against the untuned base, by contrast, the easy-data adapter is lower but not significantly so (12.4% against 16.8%, exact McNemar $p = 0.126$). The claim we can defend is the narrower one: easier data is significantly worse than harder data, not that it is significantly worse than no tuning at all.

This closes the difficulty explanation. The models can learn from calibrated data, and doing so does not improve held-out reasoning and may actively harm it.

F. The Bengali Tokenizer Tax

One finding did come out clearly in favour of the Bengali-specialised models, and it is a real one even though it did not translate into accuracy. Table V reports token counts for the same 873 training traces under each tokenizer.

TABLE V
TOKENS REQUIRED TO REPRESENT THE SAME 873 TRAINING SEQUENCES UNDER EACH TOKENIZER, MEASURED ON THE FULL SEQUENCE TRAIN_LORA.PY BUILDS: SHARED BENGALI INSTRUCTION, PROBLEM, MODEL-SPECIFIC PROMPT FORMATTING, AND REASONING TRACE. VOCABULARY SIZES INCLUDE ADDED TOKENS. NOTICE THAT THE TWO BENGALI-ADAPTED TOKENIZERS NEED ROUGHLY A THIRD AS MANY TOKENS AS QWEN3 FOR IDENTICAL CONTENT, AND THAT THIS EFFICIENCY IS NOT REFLECTED IN THE ACCURACIES OF TABLE II.

Tokenizer	Vocab	Median	p95	Max
TituLLM-1B	170,497	395	871	1,498
TigerLLM-1B	262,145	425	930	1,431
Qwen3	151,669	1,244	2,395	3,543

Qwen3 needs roughly $3.1\times$ more tokens than TituLLM to represent identical Bengali text, close to the $3.5\times$ reported by Yong et al. (2025) and consistent with the per-model premiums in Petrov et al. (2023).

The cost is concrete rather than theoretical. On a single 16GB consumer card it determined which configurations were trainable at all. Qwen3-4B could not be trained at a 4,096-token sequence limit and required a reduction to 2,048, which truncated 104 of the 873 traces. No Bengali-tokenized model faced that constraint. The same pressure appeared at inference: TituLLM-3B-Instruct pairs 3.2B parameters with a 170,497-token vocabulary, which makes the output logits tensor the binding memory cost during generation and forced a batch size of one.

It is worth noting that TigerLLM’s efficiency is inherited rather than earned. Gemma3 ships a 262k vocabulary described as balanced for non-English languages (Gemma Team, Google DeepMind, 2025), and TigerLLM adds exactly one token to it. TituLLM, by contrast, extended Llama-3.2’s vocabulary deliberately for Bengali. Both end up around three times more efficient than Qwen3 on this corpus, by quite different routes.

The broader point is that tokenizer efficiency and downstream reasoning are separable. The two most token-efficient models in this study are also the two weakest at mathematics. Efficiency on Bengali text is necessary for practical deployment on constrained hardware, and it is not sufficient for reasoning.

V. DISCUSSION

A. What the Scale Result Means

Our headline result contradicts TigerLLM’s central claim, using TigerLLM’s own released model on a Bengali benchmark. That claim was that targeted, high-quality pretraining can substitute for scale. We find that it does not, at least for mathematical reasoning: a $3,700\times$ difference in Bengali pre-training produced no measurable difference, while parameter scale produced a large and monotonic one.

This is consistent with a literature that has been converging on the same point from other directions. Kaplan et al. (2020) found performance depends strongly on scale and weakly on architectural choices. Wei et al. (2022) place the emergence of three-digit arithmetic at roughly 13B parameters and the benefit of chain-of-thought on GSM8K at roughly 100B. MathMist (Sobhani et al., 2026) reports the Bengali-English gap closing as Qwen scales. Our contribution is to separate the variables that this prior work left entangled, in the one language where the substitution claim was made.

We are careful about what this does not show. It does not show that targeted Bengali pretraining is worthless. TituLLM’s extended tokenizer delivers a real, measured threefold efficiency gain, and that gain determines what can be trained on accessible hardware. It shows that targeted depth did not buy mathematical reasoning.

B. Why Elicitation Changed Form and Not Capability

LIMA’s Superficial Alignment Hypothesis (Zhou et al., 2023) holds that a model’s knowledge comes almost entirely from pretraining and that alignment mainly teaches formatting and style. Our results are a clean, if unflattering, confirmation of exactly that. Fine-tuning taught these models to open a reasoning block, to structure a worked solution, and in TituLLM’s case to stop generating at all. It did not teach them to arrive at correct answers.

Two mechanisms plausibly contribute, and we cannot fully separate them.

The first is capacity of the adaptation method. Biderman et al. (2024) report that LoRA at rank 16, the configuration used here, reaches 0.158 on GSM8K after continued pretraining where full fine-tuning reaches 0.293, and conclude that low-rank settings substantially underperform full fine-tuning on tasks requiring new capability. The first half of that description matches what we observe.

The second half does not, and the difference is worth stating. Biderman et al. also report that LoRA forgets less than full fine-tuning and stays closer to base behaviour. Our retention data in Section IV-D shows nothing of the kind: no backbone kept even half of the items it had answered correctly, and Qwen3-4B abandoned its own output format on 222 of 250 items. We did not run full fine-tuning, so we cannot claim LoRA forgets more than the alternative. What we can say is that rank 16 on 873 examples was more than sufficient to overwrite a model’s answering behaviour, and that the common intuition of a low-rank adapter as a small, conservative edit does not describe this setting.

The second is the capability of the backbones themselves. If arithmetic competence emerges above 13B (Wei et al., 2022), then LIMO’s precondition may simply not hold in the 0.6B to 4B range regardless of adaptation method. Qwen3-4B complicates this reading, since its baseline accuracy of 44.8% shows real latent capability, and elicitation still produced nothing.

C. The Bengali Adherence Decline

The finding we did not anticipate is that training on Bengali reasoning traces made models less Bengali. Six of the seven backbones produced a lower proportion of Bengali characters after fine-tuning than before it. The decline was sharpest in exactly the models that started most Bengali: TituLLM-3B-Instruct fell from 97.9% to 80.8% and TituLLM-1B-Instruct from 98.3% to 81.3%. TigerLLM-1B fell from 87.0% to 82.7% and Qwen3-1.7B from 89.3% to 84.1%. Qwen3-0.6B was the single exception, rising slightly from 85.6% to 87.8%. Training on Bengali reasoning traces made most of these models write less Bengali, not more.

Table VI reports the paired differences with their 95% confidence intervals. In all six declining models the interval excludes zero. Qwen3-0.6B is the exception in both directions: its mean rose by 2.1 points but the interval spans zero (−1.2 to +5.4), so we do not claim it improved, even though more of its individual items fell than rose.

The likely cause is in the traces themselves. Ganit’s reasoning contains LaTeX, ASCII numerals and English mathematical notation, and models trained on it learn to write mathematics the way the traces do. This connects directly to the elicitation tax described by Qi et al. (2025), who found that forcing native Bengali reasoning costs accuracy. Our result is the same tension seen from the other side: an intervention that did not improve accuracy still moved models away from Bengali.

For work that aims at genuinely Bengali-native reasoning, this suggests language adherence should be measured and reported alongside accuracy rather than assumed. A model that improves on a Bengali benchmark by reasoning in English has not done what the benchmark intends to measure.

VI. LIMITATIONS

Adaptation method. We used LoRA at rank 16 rather than full fine-tuning. LIMO and s1 both fine-tuned all parameters. Biderman et al. (2024) document that this configuration under-acquires new capability, so our null result cannot be cleanly separated from a method limitation. We report it as a boundary on minimal-data elicitation *under the adaptation method available at consumer scale*, which is the setting most researchers in this position face.

Training duration. Models were trained for three epochs. LIMO used approximately 15 and s1 used 5. Training loss was still decreasing at epoch 3 for four of six backbones, so undertraining cannot be ruled out. No epoch ablation was run.

Architecture is confounded with pretraining depth across families. The Bengali-adapted models are built on

Gemma3 and Llama-3.2 while the scale ladder is Qwen3. Comparisons across families are directional rather than causal, as pre-registered in our study design.

One per-model deviation. Qwen3-4B was trained with a 2,048-token sequence limit rather than 4,096 because a 4B model at 4,096 exhausted the available VRAM. This truncated 104 of 873 traces, or 12%. Every other model truncated none.

Sample size is not uniform. Six of the seven backbones are evaluated on the full 250 held-out items. TituLLM-1B base is evaluated on 100, because it was dropped from the extension after producing degenerate output at 0 to 2% accuracy across its first hundred items. Every comparison is paired on the items a model completed in both conditions, so no accuracy is computed across mismatched item sets, but the Wilson interval on that one row is correspondingly wider and should not be read as tightly as the others.

The easy subset shifted source composition. Moving to the easy difficulty band raised SOMADHAN’s share from 13.6% to 31.0%, because GSM8K-derived problems concentrate there. Difficulty and source are therefore not perfectly separable in that experiment.

Training targets retain the source dataset’s markup. Ganit’s traces use `<think>` and `<answer>` tags inherited from its Qwen3 lineage. For Llama and Gemma these are ordinary text tokens, but for Qwen3 `<think>` is a control token, and combining disabled thinking mode with a target that reopens a thinking block is non-canonical. This was applied identically at training and evaluation, so the comparison is internally valid, but Qwen3 results may understate properly configured performance.

Determinism. Deterministic CUDA algorithms were not enabled, so losses reproduce to approximately three decimal places rather than exactly. Reported differences are orders of magnitude larger than this variance.

Scope. This paper addresses RQ1 only. Cross-domain transfer and confidence-based stopping were scoped out, as described below.

VII. CONCLUSION

We set out to determine whether Bengali mathematical reasoning is driven by parameter scale or by targeted Bengali pretraining, and whether minimal-data elicitation transfers to model sizes and languages outside the settings where it was demonstrated.

Scale predicts performance: within a single model family holding pretraining constant, accuracy rose from 3.6% at 0.6B to 16.8% at 1.7B to 44.8% at 4B. Targeted Bengali pretraining does not: a $3,700\times$ difference produced no significant effect. This contradicts the substitution claim made for TigerLLM, tested on that model itself.

Minimal-data elicitation produced no significant accuracy gain on any of seven backbones spanning 0.6B to 4B parameters, two pretraining philosophies and two training-difficulty regimes, including a model whose baseline accuracy shows LIMO’s precondition is satisfied. What it reliably produced

TABLE VI

BENGALI CHARACTER RATIO IN GENERATED OUTPUT, BEFORE AND AFTER FINE-TUNING ON BENGALI REASONING TRACES. Δ IS THE MEAN PAIRED DIFFERENCE IN PERCENTAGE POINTS WITH ITS 95% CONFIDENCE INTERVAL; p IS A TWO-SIDED EXACT SIGN TEST ON THE PER-ITEM DIRECTION OF CHANGE. NOTICE THAT TRAINING ON BENGALI TRACES MOVED SIX OF SEVEN MODELS AWAY FROM BENGALI, AND MOVED THEM FURTHEST IN THE MODELS THAT STARTED MOST BENGALI.

Model	Before	After	Δ (pp), 95% CI	p
TituLLM-1B (base)	90.7%	82.1%	-8.6 [-15.8, -1.5]	$< 10^{-6}$
TituLLM-1B-Instruct	98.3%	81.3%	-17.0 [-19.6, -14.3]	$< 10^{-6}$
TigerLLM-1B	87.0%	82.7%	-4.3 [-6.4, -2.3]	$< 10^{-6}$
TituLLM-3B-Instruct	97.9%	80.8%	-17.1 [-19.7, -14.6]	$< 10^{-6}$
Qwen3-0.6B	85.6%	87.8%	+2.1 [-1.2, +5.4]	$< 10^{-6}$
Qwen3-1.7B	89.3%	84.1%	-5.1 [-7.3, -2.9]	$< 10^{-6}$
Qwen3-4B	87.0%	82.0%	-5.0 [-6.4, -3.6]	$< 10^{-6}$

instead was a change in form: termination, length, formatting, and a significant decline in Bengali character adherence.

For researchers working in low-resource languages on accessible hardware, the practical implication is that the less-is-more result should not be assumed to carry down in scale. In the range most of us can train, the thousand curated examples bought presentation, not competence.

VIII. FUTURE WORK

Three directions follow directly.

Adaptation method. The clearest next experiment is full fine-tuning at the same scales, or LoRA at substantially higher rank, to separate the method limitation from the scale boundary. Biderman et al. (2024) suggests rank 256 recovers much of the gap.

Cross-domain transfer. Our original design included a second question: whether elicited mathematical reasoning transfers to non-mathematical Bengali tasks. This required the B-REASO benchmark, which was not accessible during the study period. The question remains open, and TituLLM’s own capability-specific findings (Nahin et al., 2025) suggest it is worth asking.

Adaptive test-time compute. A third question we scoped out is whether a model can judge when it has reasoned enough, using cheap untrained confidence signals rather than the calibration pipelines that current methods require. The test-time scaling literature is unsettled on whether longer reasoning helps at all, and none of it has been tested in Bengali. With a measured $3.1\times$ tokenizer penalty, the cost of unnecessary reasoning tokens is higher in Bengali than the English results suggest, which makes the question more pressing rather than less.

CONTRIBUTION STATEMENT

STATEMENT ON AI ASSISTANCE

Anthropic’s Claude was used as a coding and writing assistant throughout this project. Its contributions were: assisting in implementing the training and evaluation harness, the shared answer scorer and the statistical comparison scripts.

The research questions, the experimental design, the data curation pipeline and the selection of models and datasets are the authors’ own. Every number reported here was produced

by scripts in the project repository, and outputs were checked against raw model completions rather than accepted from summary statistics alone. No data were generated, altered or imputed by an AI system: every number reported here comes from a model checkpoint evaluated on stored held-out items, and no finding, citation or result was produced by the assistant without a corresponding computation in the repository. The authors reviewed all AI-assisted text and code, and take full responsibility for the content of this paper, including any remaining errors.

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